Mini-type Auto Washer-disinfector

Operation Manual

No.: TD-WS-B50-001 V1.14-EN

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Welcome to use small automatic cleaning washer, in order to maximize performance, please read this manual carefully before using, and strictly according to the specification requirements of installation, use and operation, to ensure that the equipment is safe and reliable service to you.

This specification applies to Smart50-T small automatic cleaning washer installation, operation and maintenance

The company produces the small automatic cleaning sterilizer takes the information as follows:

Product structure and performance: this product is mainly composed of cleaning tank, sealed door, piping system, decorative cover, control system, cleaning and other parts.

Product optimal (make) using range: this equipment is mainly used in hospital central supply room or outside the operating room's equipment (including dental), glassware, plastic utensils, cleaning, disinfection and drying.

Contraindications: product no.

Production date: see equipment nameplate. Life: 10 years or 15000 operation cycle, will be subject to first come.

Special storage and transportation conditions, methods: no.

Main sterilization factor and strength, sterilization principle: damp heat disinfection.

Kill microorganisms categories: can kill the intestinal pathogenic bacteria and streptococcus pyogenes and pathogenic yeast.

Packaging and shipping: conform to the requirements of GB/T 191.

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1. The warnings and cautions the overview

When the equipment is in operation and maintenance, must observe the following safety instructions.

"Warning" said has potential threats on personal matters, and the "notice" potentially dangerous items of equipment.To prevent these instructions () in whole or in part will be used throughout the whole explanation.Listed below are all security matters in this manual.Read the following carefully before use or maintenance equipment.

If you don't according to my company to provide the use of methods to use the equipment may damage the equipment to provide protection.

1、Equipment on the warnings of the paste, note symbol explanation:：

NO	MARK	Content
1		Surface temperature, be careful burns

2		Attention, please refer to instruction to operate
3		Grounding mark
4		Here are dangerous voltage, please power off before operation or maintenance
5		Watch your hand
6		Emergency stop button

2、In the instruction manual, use the following symbol should be noticed or attach great importance to the operation of the content.

Said should be highly valued.

Said should observe otherwise it will damage to equipment.

Said must strictly abide by, or endanger personal safety.

Warning - burns

Don't open preheating tank during the operation, water may be out.

Checking after cooling equipment for pipeline system

Warning - risk falling down

In order to prevent fall, need to keep the floor dry, dry immediately any spill liquid or condensation water dripping.

Warning - risk for personal injury or equipment damage

To operate the equipment safely and reliably, in addition to the need to adhere to this statement describes the routine maintenance, regular preventive maintenance is needed.

This equipment maintenance and adjustment should be carried out by a fully qualified professional maintenance personnel.By experience, not qualified personnel for irregular maintenance, or installation without permission of the parts, can cause personal injury.

Warning - burn shock hazard or danger

Before the repair, disconnect device of all water, electricity and gas supply.Unless all of the water and electricity supplies have been cut off in a right way,

will not allow any maintenance.

Warning - burn shock hazard or danger

The continuity of fasteners and gasket to ensure protective joint.Be sure to pack back any may be removed in the process of installation or service maintenance of gasket.

Please note anti-scald logo position of the device, so as not to burn. Cleaning and sanitation items after high temperature, please wear protective gloves to prevent burns.

Warning - danger chemical burns or eye damage

When using descaling products, should wear gloves and eye. Avoid contact with eyes or skin. In case of spillage of liquid, rinse for 15 minutes.If swallowed, do not induce vomiting. With plenty of water takes alkaline substances, immediately to the doctor for help.

Cleaning machine cleaning fluid is corrosive, exposed to adverse impact on organization, don't make it into the eyes, skin, or trying to swallow from the mouth.

Refill processing cleaning fluid, cleaning fluid container or maintenance cleaning fluid injection pump before, read and comply with the cleaning fluid precautions and instructions on the label.

When handling cleaning fluid or cleaning fluid injection pump and pipeline maintenance must wear protective gloves, masks and costumes.

NOTE:

1, this equipment meet the requirements of GB/T 18268 emission and immunity requirements.

2, this equipment according to GB 4824 class A equipment in the design and testing.Bad in the family in, the device can cause radio interference, need to take protective measures.

3, advice before equipment using electromagnetic environment assessment.

4, banned in strong radiation source (for example, the shielding of the rf source) by using this device, otherwise it may interfere with equipment to work normally.

Choosing a cleaning fluid, choose the cleaning fluid with low chloride content, high content of chloride in cleaning fluid may cause damage of stainless steel.

Cleaning equipment, use clean utensils, without grinding surface texture with the equipment in the same direction of scrubbing action.Clean appliance will damage the stainless steel grinding.Scrub with circle movement, or using a

metal brush or steel wool to wash and cabin parts can cause damage to the stainless steel door.

Install Caution：。If the equipment installed in the other side, should be able to find out one by one to close the valve and the location of the circuit breaker, in order to shut down a device at a time.

Visual inspection：

Important: a good hospital behavior specification requires that all treated by the cleaning machine equipment to visual inspection if there is a visible residue.Any visible residue instrument must pass, until clean, no visible residue may continue to final disposal.Comply with this manual instructions for use, the cleaning machine proved able to fairly smudges of the appliance is in addition to all the visible residue.

1) appliance after cooling, check each item on leave any residue or dirt.

2) any failure to properly clean objects must be deal with.

2. Product performance and technical parameters

Smart50 series small automatic cleaning sterilizer is my company recently developed a new kind of automatic cleaning and disinfection equipment, it adopts automatic computer control, the equipment structure is novel, well made, easy to use.

- Chamber size（L x W x H）mm：373x375x350
- Boundary dimension（L x W x H）mm：463x550x600（no dry）
- 463x550x750（with dry）
- Dental handpiece rack size（L x W x H）mm：363x363x214
- Instrument washing rack size（L x W x H）mm：332x127x44
- Weight： 50kg（no dry） 65kg（with dry）
- Structural style： table model
- Water consumption： 14L/cycle
- Water heating mode： electric heating
- Outfall： Φ25.4mm
- Exhaust port： Φ25.4mm（no dry）
  Φ50mm （with dry）

- Water processing temperature: 93℃
- Water source pressure: 0.2～0.5Mpa pure water or soft water
- Water source connector: 1/2″ external thread
- Soft water: 4-7ºd（1 ºd=0.178mmol/L）
- Pure water: 0°d

1ºd=10mg / L cao = 17.80mg / L $caco_3$

Note: if the user softened water can not meet the above requirements, recommend the use of distilled water or deionized water instead of demineralized water, otherwise easy to cause the instrument with a white spot.

Note: customers will provide sewer system for sewage trap tube for water seal, and can be removed for cleaning, or port of fittings can be easy to clean。

★Power：AC 220V±22V,, 50Hz±1Hz ，3.5kva

●Using the environment：

- Environment temperature：5-40℃
- Relative humidity：≤80%
- Place：

A. No magnetic field around;

B. Avoid high temperature and high humidity;

C. Without direct sunlight;

D. Around without fire;

E. Maintain equipment level natural place;

F. Avoid heavy dust, oil mist, corrosive gas and combustible gas;

G. Good natural ventilation.

2.1. The main electrical components power

Circulating pump	0.3 kVA
Air heating tube	1.5 kVA
Draught fan	0.3 kVA
Water heating tube	3.0 kVA
Control system	0.2 kVA

3. Installation and debugging

3.1. Regular safety standards

The equipment must be according to the installation instructions to install;

Must by specially trained personnel to install and maintain equipment;

Prior to servicing and maintenance of the equipment, the operator must read carefully the instructions;

If there is a leak equipment, must be timely maintenance;

Devices cannot be water splash;

The use of corrosive cleaning fluid, must be careful;

The equipment must be grounded, to prevent accidental leakage.

3.2. Equipment transportation and unloading

3.2.1.Transportation

Equipment to prevent collisions in the process of transportation, be affected with damp be affected with damp, and the effects of the weather.

Under no circumstances can equipment inclination flip.

If possible, transfer equipment used forklift fork, if you must use the crane, so, crane cable and cable can not have pressure on the equipment the outer anywhere.

3.2.2.Unloading

Before unloading equipment, please read carefully the instructions.

Equipment unloading, installation, connection and put into use a professional must guide.

If the device is stored before using, before turning on the power supply shall ensure that the equipment 24 hours exposure to remove moisture.

3.3. The power cut off the device

In the power supply line, you must install a separate power switch, to cut off the power supply.The wall of the switch should be installed near the equipment, and convenient operation;To distinguish power to cut off the device id in order to avoid confusion.

3.4. Installation

Equipment installation, should be guided by professionals, by construction professionals.

Low with forklift or other operating tools will equipment shipped to the settlement;

Put to move the equipment, adjustment of the four bolts, make cleaning tank is in a state level;

Note:

1) in the process of moving, don't put the decorative cover damage or scratches.In case of special circumstances, should be under the guidance of professional staff, will cover torn down, after the equipment is in place, as is restored.

2) installation, make sure equipment in horizontal.

3), when installation should be careful, the equipment must be accurately adjust level, can be adjusted by adjusting the equipment of the four bolts, to adjust the level.

4) do not use equipment in difficult to disconnect the location of the device operation

5) installation environment:

Avoid installed in heavy dust, oil mist or containing conductive particles, corrosive gas, flammable gas environment.

Avoid installed in place of easy by electric shock or vibration.

Avoid installed in high temperature and high humidity or places easily by the rain.

Avoid installed in a strong magnetic field environment.

Avoid installed in the place of direct sunlight.

Keep the equipment level natural place.

3.5. The equipment of water, electricity, drainage of connection

Note: must be installed by a professional or guidance

According to the installation drawing and installation requirements, water, electricity equipment to identify the corresponding position.Filling material should be used in pipeline tapping seal to prevent leakage.

3.5.1.Electrical connections

- Near requirements in equipment installation on the wall of the power switch box, switch box must be installed in a single-phase circuit breaker, breaker rating to 20 a.
- Switch on the wall is suitable position, make the operator can easily operation.
- In order to ensure the safety of the person, equipment, must be equipped a ground wire, control cable a mustard yellow green line must be connected to the ground wire is reliable, set line and the line should be horizontal flat vertical and effective fixation.

3.5.2.Water connection

Note: if the equipment installed in the other side, should be able to

find the shutoff valve

Door and the position of the circuit breaker switch to shut down a device at a time

★ If the water pressure is lower than 50 kpa, must add a booster pumps.

Install on the road u feed line filters.

3.5.3.Drain connections:

- According to equipment drain mouth size, to guide a drain alone in the gutter, gutter of displacement is not less than 1 minute 50 litres, specific size reference equipment installation figure
- 1.1 function
- Electrical equipment, such as water, before the electric equipment, carefully check, see the connections between whether fastening equipment, no loose, whether each line pipe clamp joint fastening, whether fastening screw joint, if there is loose, please tighten it.
- Carefully check, confirms the accuracy of equipment wiring, and into, drain trouble-free, open water switch.
- Pour into a corresponding plastic bucket with a cleaning fluid, and in the liquid level detection device.
- To ensure that the cleaning hold water has reached the set water level.

4. The structure and composition

4.1. Cleaning tank

- Cleaning tank by the high quality mirror stainless steel, treatment is excellent;
- Using spot welder welding, reduce the welding deformation;
- Clear a tank internal injection arm: can jet stream, all-round cleaning items;
- External surface cleaning module with good performance of insulation insulation materials, can reduce heat loss, reduce noise.

4.2. Seal the door

Sealed door selects the stainless steel material, the overall elegant appearance, design structure is easy to operate.

4.3. The piping system

- Large flow solenoid valve to realize the function of automatic cleaning hold water;
- Import and pump to realize the function of automatically into the cleaning agent,

and can be set according to the procedure, control cleaning agent into the quantity;

- Imported large flow circulation pump, circulating water by spraying arm, rotating flow, to effective cleaning of objects, and on the tube and bottle class objects in internal cleansing.
- In different stages, according to the procedures set, automatic control, electric heating tube for cleaning in circulating water heating, at the temperature of need;
- Disinfection in circulating water can be heated to 93 ℃ and maintain, for cleaning objects in thermal disinfection.
- After each cleaning water automatic discharging, lest produce items pollution;

4.4. The decorative cover

By the refined from imported stainless steel wire drawing board

4.5. Control system

Working process for washing, cleaning, a rinse, secondary rinse, disinfection, drying.

30 sets of editing program, users can according to the need for custom parameters.

Controller is integrated with control, display, intuitive interface, easy to operate.

4.6. Washing rack

This equipment can accommodate a dental cleaning frame, a equipment cleaning basket, equipped with double jet arm.Also optional in other special special cleaning, such as glass cleaning, equipment cleaning, etc.

4.7. Loading and unloading

Using artificial to loading and unloading.Form a complete set of cleaning basket frame and may need to clean all the equipment in accordance with the principle of ergonomics, properly placed.

4.8. Safety device

This equipment use body expansion thermostat as chamber heating tube prevent dry heating protection device, can prevent dry heating tube chamber.

5. Control panel operation

5.1. Program operation

Control panel：From left to right in turn is the text display operation, stop, up and down.Control panel has a POWER button on the right, as the equipment POWER

switch, inside the electrical box have a main switch, equipment installation in place and the switch has been open.

Devices power on, text display "figure 1" into the interface

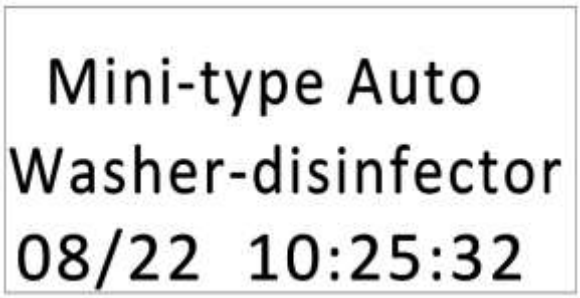


Figure 1

Under this interface, press the up or down key, can run respectively in the process of 01-02 or edit selection interface, when press the up key,into "figure 2" interface

Figure 2

Under "figure 2" interface, through up and down button can move the cursor, elected in the "RUN", press the "RUN" button, the program runs, "figure 3" into the interface, was elected in the "edit", click "RUN" button, into "figure4" interface.

Figure 3

"Figure 3 displays the application stage, status, the total run time and disinfection A0 value, T1 for the chamber temperature.

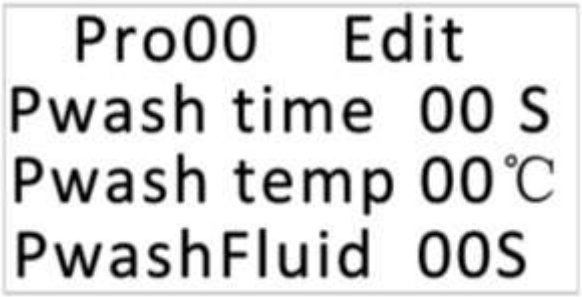


Figure 4

In figure 4, you can set the program in different stages of the operation parameters, can be set beforehand drainage time, temperature and washing time and washing rinsing time, disinfection, sterilization temperature, etc., and can according to need to close the rinse 2.Move through the "RUN" option, increase and decrease key set operating parameters, the key to increase the parameter value, the increasing of key reduce parameter values.Set up is completed, press the "STOP" button, exit the program editor interface, the system can automatically save the set parameter values.

5.2. System settings

Under "figure 1" interface, press the "STOP" button for 5 seconds, enter the system Settings interface, as shown in figure 5 ",

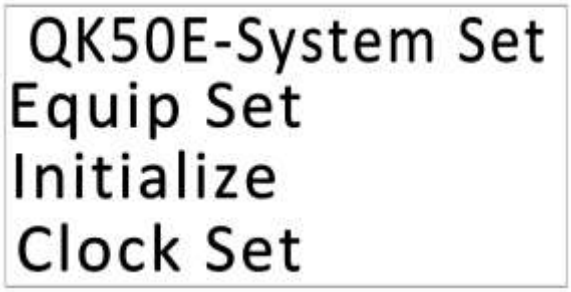


Figure 5

In the "5" screen, by up and down key mobile options, click "RUN" button to enter

5.3. Clock calibration

Figure 6

Under the interface, "RUN" key to move the cursor, when a certain parameter

values are in a state of flashing, at this time can adjust the parameter value by increasing and decreasing key, keys to increase the parameter value, the key to reduce reduction value, set up is completed, press the "STOP" key to exit.

5.4. Factory parameter

Back to the system default parameter value, press "RUN" button for the factory parameters setting

5.5. Equipment maintenance

Choose equipment maintenance function, into "7" interface

Figure 7

Input password "11" click "RUN" button to enter the "8" figure interface

Figure 8

5.5.1.The input test

Choose "input in" 8 "figure interface detection", press "RUN" button to enter "figure 9" interface, when the input signal, shown as a solid point

Figure 9

1 chamber water 2 cleaning liquid level 4 door shut

5.5.2.Manual operation

In "8" figure interface to select manual operation, select manual operation, press the "RUN" key into "figure 10" interface

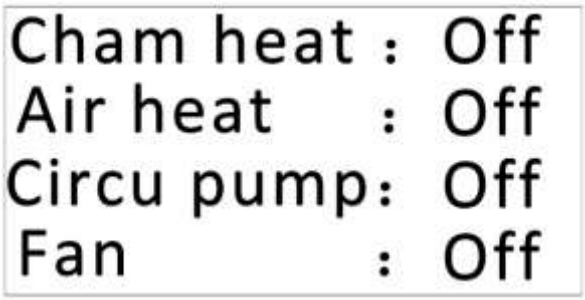


Figure10

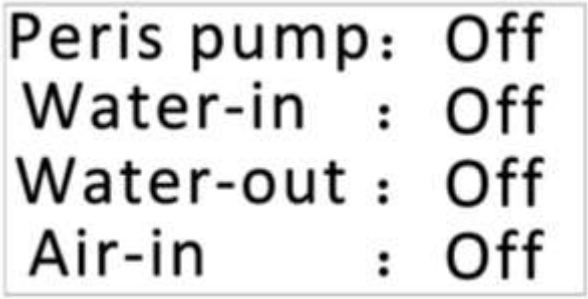


Figure11

By up and down key mobile options, "RUN" to open the part, according to the "RUN" button again, close the part.

J1 Chamber heating tube J2 Air heating tube J3 Circle pump J4 Fan

J5 peristaltic pump J6 Water inlet valve J7 Water outlet valve J8 Air inlet Valve

Press the "STOP" button to exit the manual operating interface, return to the interface at the next higher level, and all parts manual operation to STOP work.

Note: the chamber water level has not been achieved, the manual heating could not be opened.

5.5.3. Deviation set

Choose "deviation in the" 8 "figure interface Settings", into the figure 12" interface, through "the RUN button set items, up keys to increase the parameter value, the down key to reduce reduction value.

Figure12

6. Application process instructions

6.1. Program introduced

System Settings 01-07, a total of seven preset program, respectively is:

01 surgical instruments

02 ware instruments

03 plastic products

04 transmission items

05 severe pollution

06 cleaning procedures

07 disinfection procedures

08 drying procedure

Note: plastic products program default setting is 85 ℃, please make sure the disinfection plastic products can withstand this temperature;Glassware and plastic devices to use the corresponding cleaning, so that we can make the equipment inside and outside surface temperature is consistent, namely cleaning to 45 ℃, the sterilized at 93 ℃ (instruments) allowed by the temperature, and glass and plastic products used by the connector size of Φ 10 mm, can fully achieve spray effect.

Surgical instruments and dental are suitable for surgical instruments, so you can use this two kinds of load at the same time put in washer disinfector in cleaning.

6.2. Brief introduction

1) Prewashing（if have）

Start, first of all to drainage of 45 seconds, then fill valve open, clean cabin started into the water, and after the detection of water level, circulating pump open, delay stop after 6 seconds into the water, the water valve closed, start time, cycle cleaning circu-

lation pump after 2 minutes (time can be set) closed.Drain valve open, began to drainage, 60 seconds after the discharge valve closed, turn into the next phase.

2) Washing process

Inlet valve is opened, the cleaning chamber started into the water, the peristaltic pump, open the cleaning fluid into the cabin, after the detected water level, circulating pump open, delay stop after 6 seconds into the water, the inlet valve closed, hold open cycle heating heating tube.Cleaning cabin temperature rose to 45 ℃, temperature can be set, electric heating tube closed, stop heating, start time, cycle cleaning circulation pump after 6 minutes (time can be set) closed.Drain valve open, began to drainage, 60 seconds after the discharge valve closed, turn into the next phase.

3) First rinsing process

Inlet valve is opened, the cleaning chamber began with water, after the detected water level, circulating pump open, delay stop after 6 seconds into the water, the inlet valve closed, and start timing, timing 60 seconds (set), circulating pump closed, drain valve open, began to drainage, 60 seconds after the discharge valve closed, to get to the next stage.

4) Second rinsing process

Inlet valve is opened, the cleaning chamber began with water, after the detected water level, circulating pump open, delay stop after 6 seconds into the water, the inlet valve closed, and start timing, timing 60 seconds (set), circulating pump closed, drain valve open, began to drainage, 60 seconds after the discharge valve closed, turn to the next stage.

5) Disinfect

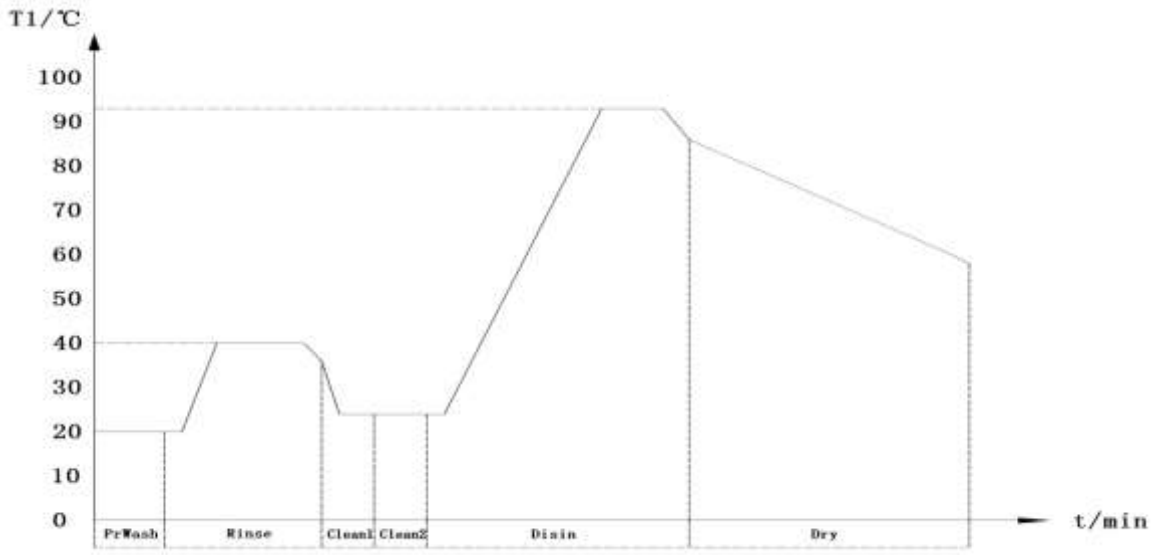
Inlet valve is opened, the cleaning chamber began with water, after the detected water level, circulating pump open, delay stop after 6 seconds into the water, the inlet valve closed, cabin heating, when cleaning tank temperature up to 93 ℃, temperature can be set, electric heating tube closed, stop heating.Start timing, circulation cleaning after 150 s (set), circulating pump closed, drain valve open, began to drainage, after 60 seconds, transferred to the next stage.

6) Dry

Into the air valve open, delay after 5 seconds, the fan and the heating tube open, cold air by the fan through the electric heating tube heating after blowing into the cleaning chamber, fan open 15 seconds after the discharge valve closed, when the outlet

temperature reaches set after drying temperature, heating tube disconnected, insulation, within the range set point up and down 5 ℃ drying time arrival, heating tube to stop working, and fan cooling after 60 seconds, the buzzer alarm, press the stop key to exit.

6.3. Procedure chart



T1：chamber temperature

6.4. The program is running process parameters

1) the cleaning and disinfection of the maximum rate of change of temperature is 3 ℃ / min.

1) Equipment chamber allows maximum temperature of 100 ℃.

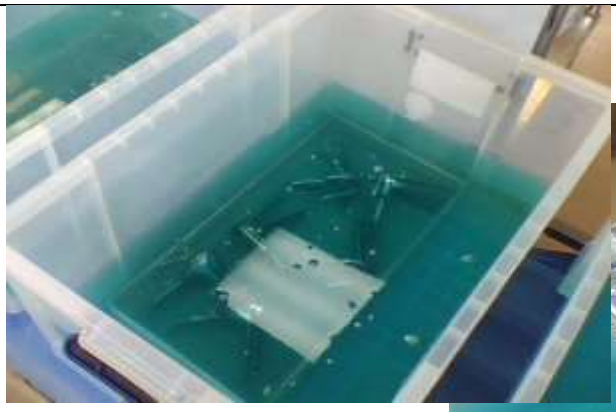
2) Equipment from cold start to run a cycle is about 32 minutes, equipment from hot start running a cycle takes about 30 minutes.

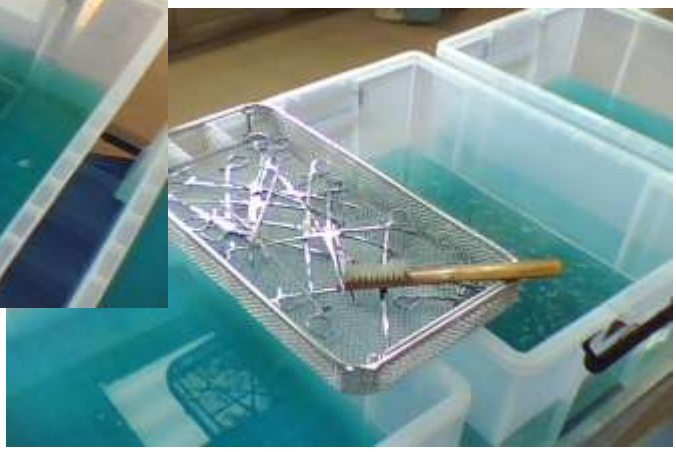
3) Wet and heat sterilization process, the coldest place for equipment load the top right corner, the temperature is about 93 ℃;The hottest place for device outlet, load temperature about 94 ℃.

7. The cleaning steps

7.1. Pretreatment

Surgical instruments, dental instruments, utensils, and the glass pretreatment were put inside preprocessing trough soak 5-10 min





7.2. Opened the door of loading

Equipment door gate of manual operation, the thumb holding down the button on the door, the rest of the fingers hold the hand, the thumb press the button, outward sliding door, can be opened, after opening the loading door horizontally state.

7.3. Loading

Choose the appropriate cleaning load, depending on the type of load this sterilizer can be equipped with the following cleaning.

1) Dental washing rack

Dental cleaning small automatic cleaning sterilizer for my company standard products, mainly for dental cleaning and disinfection of the mobile phone or similar device.Dental cleaning rack can be placed 18 dental mobile phones.In addition small automatic cleaning sterilizer standard configuration

An instrument basket, supporting the use, is mainly used for holding forceps, scissors, tweezers and other surgical instruments.

Operation, dental mobile phones directly into the dental cleaning can be shelf, put into the cleaning tank cleaning, close the door to run the program.

Dental washing rack

2) Instrument rack

Small automatic cleaning sterilizer is standard equipment cleaning, the cleaning frame includes four instruments basket, it is mainly used for serving surgical instruments on pliers, scissors, tweezers, etc.

Instrument rack

7.4. Loading door

Manual closed door, after the sound heard the door hook, said the door has been locked in place.

7.5. Program selection

Depending on the type of wash load selection process, and the applicable scope of the various procedures default arguments are as follows:

Program 01 surgical instruments: suitable for surgical instruments and dental cleaning and sanitation.

Program default parameters:

Prewashing time	**Prewashing temperature**	**Liquid inlet**	**Wash time**	**Washing temperature**	**Liquid inlet**
0	0	0	300s	40℃	6s
Rinse time	Disinfect time	Disinfect temperature	Dry temperature	Dry time	Cooling time
60s	150s	93℃	110℃	15min	60s

Program 02 vessels instrument: mainly applies to a beaker, jars, tubes, triangle cup of vessels laboratory in cleaning and sanitation.

Program default parameters:

Prewashing time	**Prewashing temperature**	**Liquid inlet**	**Wash time**	**Washing temperature**	**Liquid inlet**
0	0	0	300s	40℃	6s
Rinse time	Disinfect time	Disinfect temperature	Dry temperature	Dry time	Cooling time
60s	150s	93℃	110℃	15min	60s

Program 03 plastic products: suitable for plastic products of cleaning and disinfection

Program default parameters:

Prewashing time	**Prewashing temperature**	**Liquid inlet**	**Wash time**	**Washing temperature**	**Liquid inlet**
0	0	0	480s	40℃	6s
Rinse time	Disinfect time	Disinfect temperature	Dry temperature	Dry time	Cooling time
60s	1000s	85℃	100℃	20min	60s

Program 04 severe pollution: applicable to high levels of pollution item of cleaning and sanitation.

Program default parameters:

Prewashing time	Prewashing temperature	Liquid inlet	Wash time	Washing temperature	Liquid inlet
120s	0	0	480s	40℃	10s
Rinse time	Disinfect time	Disinfect temperature	Dry temperature	Dry time	Cooling time
60s	150s	93℃	110℃	15min	60s

Transmission program 05 items: apply preliminary infection item of cleaning and sanitation.

Program default parameters:

Prewashing time	Prewashing temperature	Liquid inlet	Wash time	Washing temperature	Liquid inlet
120s	0	0	300s	40℃	6s
Rinse time	Disinfect time	Disinfect temperature	Dry temperature	Dry time	Cooling time
60s	150s	93℃	110℃	15min	60s

Program of 06 cleaning: separate items to cleaning processing.

Program default parameters:

Prewashing time	Prewashing temperature	Liquid inlet	Wash time	Washing temperature	Liquid inlet
0	0	0	300s	40℃	6s
Rinse time	Disinfect time	Disinfect temperature	Dry temperature	Dry time	Cooling time
0s	0s	0℃	0	0	0

Program 07 disinfection: will cleaning disinfection treatment alone, need to cleaning items then use this program for disinfection.

Program default parameters:

Prewashing time	Prewashing temperature	Liquid inlet	Wash time	Washing temperature	Liquid inlet
0	0	0	0s	0℃	0s
Rinse time	Disinfect time	Disinfect temperature	Dry temperature	Dry time	Cooling time

0	150s	93℃	0	0	0

Program 08 dry: on the drying processing.

Prewashing time	Prewashing temperature	Liquid inlet	Wash time	Washing temperature	Liquid inlet
0	0	0	0s	0℃	0s
Rinse time	Disinfect time	Disinfect temperature	Dry temperature	Dry time	Cooling time
0	0	0s	110℃	15min	60s

7.6. Run the program

Choose good programs, run the program. Observed whether equipment alarm when running the program, such as leaky anomalies, if appear this kind of phenomenon, should immediately exit, close the equipment power supply, contact your local customer service personnel for repair.

7.7. Door open

Program ended, the thumb holding down the button on the door, the remaining fingers hold the clasp hands, press the button, the thumb to the sliding door, can be opened, after opening the loading door horizontally state.

7.8. Remove the sterilized items

Pull the wash rack cleaning tank to seal the door, for items after cooling, take out the items.

8. Equipment alarm and processing method

No.	Information	Descriptions	Analysis and solutions	Triggering phase
1.	Please close the door before running!	The system cannot detect the signal of door close position.	1. Re-close the door and confirm the alarm. 2. Replace the door close position switch and rewiring. 3. Adjust the place of door close position switch.	Run phase
2.	Water injection slow!	Water cannot reach certain level within certain time.	1. The water pressure is low. 2. The water valve is not opened. 3. Replace the water injection valve. 4. Water injection valve wiring failure.	Prewash, wash, rinse, disinfection
3.	Electrical air heating tube protection!	The protection switch of air heating tube has detected signal.	1. Drying tank temperature is too high. 2. The setting value of protection switch is too low. 3. The protection switch is broken. 4. Fan is not working or blast pressure is too low.	Dry
4.	Electrical chamber heating tube protection!	The protection switch of chamber heating tube has detected signal.	1. The temperature of chamber heating tube is too high. 2. The setting value of protection switch is too low. 3. The protection switch is broken.	Prewash, wash, disinfection
5.	Blast pressure low!	Blast pressure switch has not detected pressure.	1. Check whether the fan is working normally; check whether there is any air leakage. 2. Check whether the pressure switch is broken; Check whether the setting value of switch is too high.	Dry
6.	Please inject cleaning fluid!	The level floater has not detected signal.	1. Lack of cleaning fluid, please inject. 2. Check the floater switch.	Any
7.	Drying system heating slow!	The heating speed of drying system is lower than the default value.	1. Check whether the air heating tube is broken. 2. Check the heating tube wiring.	Dry
8.	Chamber heating slow!	The heating speed is lower than the default value.	1. Check whether the chamber heating tube is broken. 2. Check the heating tube wiring.	Prewash, wash, disinfection

No.	Information	Descriptions	Analysis and solutions	Triggering phase
9.	Dry heating pipe protection!	Heating pipe protection switch signal is detected.	A, drying oven temperature is too high B, protect switch Settings are too small C, if there is a damage protection switch	Any
10.	The water in chamber is not discharged totally!	System has detected water level after pre-drainage stage.	1. Check whether the drainage valve is broken. 2. Check whether the drainage pipe is blocked.	Prewash, wash, rinse, disinfection
11.	Temperature probe of drying system is broken or not connected!	The temperature is higher than 150℃ or lower than 0℃.	1. Check whether the probe is broken. 2. Check the temperature probe wiring.	Any
12.	Chamber temperature probe is broken or not connected!	The temperature is higher than 150℃ or lower than 0℃.	1. Check whether the probe is broken. 2. Check the temperature probe wiring.	Any
13.	Tuyere water level alarm	Detect water inlets when equipment operation process	A, check the air inlet valve is damaged. B, testing water inlets, float is damaged. C, contact the engineer	Any

9. Common failures

NO	Failures	Handling method 法
1	Inflow slow	★ Check water electromagnetic valve is blocked ★ Check if water pressure is too low ★ Contact the engineer
2	Inner chamber heating up slowly	★ Check whether the heating element is damaged ★ Check whether the voltage is normal ★ Contact the engineer
3	Drainage time is too long	★ Check whether the drain valve is damaged ★ Check the drainage pipe is blocked ★ Contact the engineer
4	Temperature sensor failure	★ Check whether PT100 failure ★ Contact the engineer

10. Routine inspection

10.1. Routine inspection

- Every time after the program is running, check for item drops in internal filter

module, and if there is must be removed in a timely manner.

- Check whether the spray arm is able to flexible rotation, check whether there is injection arm holes blocked;
- , if necessary, you can clean spray arm off in the middle of the spray arm cover (or fittings and accessories), remove the spray arm and rinse after reinstall check whether injection arm rotate freely.
- Cleaning cleaning tank at the bottom of the filter;
- Every day before running the program for the first time, check whether there is cleaning liquid inside the cleaning barrel;
- Check and cleaning tank cleaning rack inlet on the water inlet of accurate connection, if the connection is not accurate, timely maintenance.

10.1.1.Chamber interior maintenance

Use the acid solution to clean up the scale on the chamber inner surface. Wash the chamber by spray arm repeatedly.

10.1.2.Chamber exterior maintenance

1) The following procedures are used for cleaning the exterior of chamber.
2) (1)Dip the wet cloth or sponge into the cleaning detergent to wipe the chamber outer surface along the chamber texture.
3) (2)Wash off the cleaning detergent completely.
4) (3)Use dry and clean cloth to wipe the surface.

10.2. Regular maintenance

The frequency of equipment maintenance depends on the water quality and operation frequency. We recommend the following methods:

Parts	Frequency				
	Check yearly	Replace base on actual situation	Replace yearly	Check Weekly	Replace in every 1000 hours
Sealing gasket	●	●			
The pipe between the peristaltic pump and detergent bottle		●	●	●	

Parts	Frequency				
	Check yearly	Replace base on actual situa-tion	Replace yearly	Check Weekly	Replace in every 1000 hours
The pipe between the peristaltic pump and the equipment		●	●	●	
The pipe between the blast pressure switch and fan		●	●	●	
All silicon gasket		●	●		

Besides the above regular inspection, it is necessary to carry out the yearly maintenance.

Function inspection

(1)Circuit and switch

Make sure that the switch works well and the circuit is undamaged.

(2)Filter and valve

Make sure that the valve and switch of detergent level detecting are working well.

Check and clean the filter in the water injection pipe.

Check the connection of all the pipelines.

(3)Control system

Check whether the buttons on the equipment are working well.

(4)Sealing door

Make sure the sealing gasket is undamaged and there is no air or water leakage. If necessary please clean or replace the gasket.

(5)Metering pump

Make sure the sucking pipe and metering pump are filled with detergent before program starts.

(6)Washing system

Make sure the spray arms can rotate well.

Make sure the nozzles are not blocked. Please clean them if necessary.

Make sure the filter which is on the chamber bottom has been installed properly.

Make sure the connections of washing system, pump and chamber are reliable.

Make sure all the silicon and plastic parts are undamaged.

(7)Drying system

Check the pipeline system once a year.

Check the overheat protection device at least once a year.

10.2.1.Washing system

At work, all the internal water interface must be closely

- Confirm injection arm can free rotation.
- Confirm that injection arm hole without congestion, otherwise, according to the need to clean.
- Confirm the clean room at the bottom of the screen pack installed correctly, and according to the need for cleaning.
- Confirm the cleaning system connected to the pump and the chamber fastening, according to the need to tighten and effective seal.
- Check all rubber and plastic parts is not damaged.

10.3. Maintenance and replacement of parts

All maintenance and replacement of parts need to cut off water and power supply, under the premise of otherwise danger!

10.3.1. Replace the peristaltic pump

- If there are one of the following situations, need to replace the peristaltic pump:
- Peristaltic pump, accurate, but the motor is still not working.
- Motor drive failure.
- The amount of liquid is not accurate.

Replacing peristaltic pump need to wear protective gloves to prevent chemical liquid hand injury.

Peristaltic pump line to change once a year, so the line change is very important, first remove the screw, remove the can replace pipe lines.

10.3.2.Sealing gate level

If seal is not level and compare the obvious, must be adjusted.

Tear down the equipment in the cover cover (left side), after both side in a sealed door can see two M6 adjustment bolt, door height can be adjusted by adjusting the

bolt seal levels (when doing this adjustment, seal is best under the conditions of lift door, easy to operate at this time).

10.3.3. Adjust sealing effect

1) If door sealing effect is not good (circulating pump seal door while working position have water leakage), you will need to adjust the sealed door.
2) 1) the first check whether the seal is broken cause leakage, or cleaning tank welding crack caused by the leak, if not for the following;
3) 2) remove the roof leak sealing door cover, control, and the left and right sides of the cover, the first order open cover cap, after control cover, left and right side cover;
4) After 3) cover the top with four bolts and cap is linked together, to ensure the equipment remove the bolts have been removed, at this time only need to remove the cap, push forward hood, which can be scored hood, removed, control the hood in the silicone cap hexagon screws, mobile control on the cover, under cover can take control, open her left the right side of the cover and the bolt below, can win left and right side cover;
5) 4) you can see at this time to install the two side of the sealed door board, through the round hole and equipment body are connected together, loosen all fixed screw, adjust the sealing door and clean the tank sealing ring relative position, until can achieve good sealing effect.
6) 5) installed side cover, the control module in the reverse order cover, top cover, back cover.

10.3.4.Detection of replace seal door closed switch

1) If sealed door closed detection switch is damaged, must be replaced
2) 1) remove the mask, cap, after;
3) 2) at this time you can see the door shut a detection switch, replace or adjust the good location, good cap, control cover.

10.3.5. Replace the door sealing strip

1) If the door sealing strip is mechanical damage, crack or the door itself has a leak, it must be replaced.
2) 1) open the sealed door;
3) 2) remove the sealing strip, sealing groove sealant to clear clean;
4) 3) back along the bottom of the seal groove on a layer of sealant, 2 to 3 hours to reload the new sealing strip;

5) 4) whether the inspection door seal.

10.3.6. Replace the drain valve

If the relief valve body and drainage is broken, need to replace the drain valve.

If electromagnetic valve damage or does not work, the need to replace solenoid valve, the remove the top cover, back cover, left and right side cover, control cover, remove the solenoid valve line pipe clamp, can replace the solenoid valve.

10.3.7.Replace the water tank temperature probe

1) If the controller on the prompt "xx probe failure", it must be replaced after confirmation are correct
2) 1) to remove the top cover, back cover, left and right side cover, control cover (method) 3, wherein, the temperature probe in the bottom of the tank cleaning, heating tube side;
3) 2) remove the temperature probe attachment;
4) 3) loosen the tighten nut probe, take away the old probe;
5) 4) the new probe carefully into module;
6) 5) check the probe function.

10.3.8.Replace the water tank temperature probe

1) If the controller on the prompt "xx probe failure", it must be replaced after confirmation are correct
2) 1) to remove the top cover, back cover, left and right side cover, control cover (method) 3, wherein, the temperature probe in the bottom of the tank cleaning, heating tube side;
3) 2) remove the temperature probe attachment;
4) 3) loosen the tighten nut probe, take away the old probe;
5) 4) the new probe carefully into module;
6) 5) check the probe function.

10.3.9.Replace the circulating pump

If the damage or leakage of the shaft seal is circulating pump motor, it must be replaced.

If the equipment in use process, the circulating pump for some situation is damaged, need to replace or repair cycle pump, you need to remove the circulation pump, the situation is in the outer garment, tailgate (on both sides by four bolts fixed on the bracket), in addition to seal the door cover without dismantling, com-

plete dismantling of circulating pump, first disconnect the circulation pump mounting plate fixed bolt, and line pipe clamp (each line need to pull down a pipe clamp), and then disconnect the power cord, the circulating pump has been removed, from the back of the can,

Change new circulating pump, circulating pump in the reverse order of dismantling mount, according to the wiring instructions can be connected to the power cord.

10.3.10.Repair the rotating arm

1) If spraying arm shaft deformation or arm to wear, it must be repaired.
2) Operation:
3) 1) unscrew the center of the rotating arm cover, or the fittings and accessories;
4) 2) removed from the shaft injection arm;
5) 3) injection arm to the cause of the problem.

10.3.11.Change inlet valve

1) 1) the equipment at the back of the inlet valve is installed in equipment, after removed, you can see green inlet valve;
2) 2) from the set back to replace the damaged water electromagnetic valve;
3) 3) connected the power cord to check water electromagnetic valve function.

10.3.12.The gate leakage repair equipment operation

Equipment operation stage gate leakage water the following conditions:

Cleaning phase leakage;Rinse phase leakage;Disinfection phase leakage.

a. Cleaning phase leakage

1) If found leaking started running cleaning stage, suspend the program first, open the door, water level of observation, if the water level is higher than the door strip installed at the bottom of the tank, may be the water level is too high, lead to equipment leaking, adjust the water level to the appropriate location.
2) Water level control methods:
3) (1) the program to adjust, adjust the water time this method is generally applicable to professional).
4) 2) water pressure adjustment, within the scope of the equipment requirements of the water pressure increase, reduce the water pressure, this method is suitable for the environment for water supply equipment, water pressure can be adjusted).
5) 3) machine to adjust, adjust the equipment near the door of the two machine

foot fixed machine (the latter two feet), can adjust equipment into the water, raise the equipment into the water to reduce machine feet, lower equipment into the water to increase machine feet, (adjust the notice about equipment level (in the face of equipment), and the equipment machine in front of the foot (near the door side) should not be lower than the machine equipment foot (away from the door side)).

6) 4) adjust float, disconnect the equipment after the cover, you can see water float, raise the float device, the water cut float less equipment into the water.

If the water inside the equipment to moderate

- Maintenance method one: test washing machine drain (exhaust), whether the line is too long (more than 2 m), appropriate cut short line.
- Inside the door in the pipeline water, clear water in the pipeline, rearranging line, make it not water;
- Pipe is clogged, if the plug is rearranging intersection;
- If the above method or leaking, observation door sealing strip,
- If the strip to install good, no damage, properly adjust the door sealing strip highly,
- If door sealing strip damage must replace the door sealing strip.
- A. Rinse stage is leaking
- Suspend the program after the first observation of water level, if the water level is low, as a maintenance repair methods mentioned above,
- If the high water level observation device first internal clean whether there is a large number of vacuoles, if has the need to adjust the equipment into the cleaning time, and then press the steps in the operation manual, into the custom program into the cleaning fluid inside the equipment is zero, adjust time run to a second rinse and drainage equipment, exit the program.
- If cleaning vacuole is moderate, according to the above regulation water level method of regulating water level.

b. Disinfection phase leakage

Leakage according to the maintenance method of a test can maintenance at this time.

10.3.13.Detergent to add and change

If the equipment appears "timely supplement cleaning fluid is running out, please!"When the alarm, need to replace the cleaning fluid at this time.

First remove the cleaning fluid buoy, replace the cleaning fluid

After the replacement, and then cleaning fluid buoy.

10.3.14.Replace the silicone hose

1) Regularly inspect rubber hose if damaged or aging, please replace the silicone hose.
2) The operation process
3) 1) unscrew fastening pipe clamp;
4) 2) remove the silicone hose;
5) 3) new silicone hose;
6) 4) fastening pipe clamp;

10.3.15. Replace the fuse (if any)

1) If equipment failure cause damage of fuse.
2) The operation process
3) 1) will power equipment;
4) 2) remove the damage fuse;
5) 3) change the new fuse;

10.4. Main parts list

No.	Name	Type	QTY
1	chamber	Smart 50.01	1
	door	Smart 50.03	1
	Limit switch	D4D series	1
2	Mian board	SMT175B	1
	screen	SMT50A	1
	Temperature sensor	0～180℃	2
4	pipeline	Smart 50.04	1
	Circulating pump	Hanning series	1
	valas	Fseries	1
	Dosing pump	811 series	1
	Draught fan	G-200 series	1

10.5. Accessories list

No.	Name	Type	Remark
1	Rack	14680-003.05.00.R3	
2	Cleaning basket	14680-003.05.00.R1	

11. Appendix

11.1. Operating procedures

Small automatic washer operating procedures

Preparation:

1) Closed equipment total power switch, gently press equipment on the power button.
2) Check whether there is water cleaning chamber, and water is clean the water manually.
3) After open the power switch, such as display cleaner is insufficient, add detergent to plastic, detergent.
4) Each injection arm ø opened the door, manual revolving, such as rotation is not flexible, to adjust.

Operation:

1) Before u program is running, manually open the door, door open after the items from the sealed door into.
2) After the door closed to run the program, in the process of running the program can't open the door, prevent burns.
3) Goods display to ensure that the goods will be able to get a thorough cleaning, pay attention to the items don't hinder the rotation of the hydraulic arm.
4) During operation, the operator shall not be far away from the equipment, should closely observe the equipment running status, if there are any abnormal, timely processing, to prevent accidents.
5) During operation, there should be no serious leak phenomenon.
6) After the program, manually open the door, open the door after items out from under the board, be careful not to burn.

After work,

1) Gently press equipment on the power button, to cut off the equipment power supply.
2) Closed water ¨.
3) Finished daily work, cleaning tank should be kept clean, inside and outside board dirt should be cleaned clean, once a week for small maintenance, monthly maintenance.

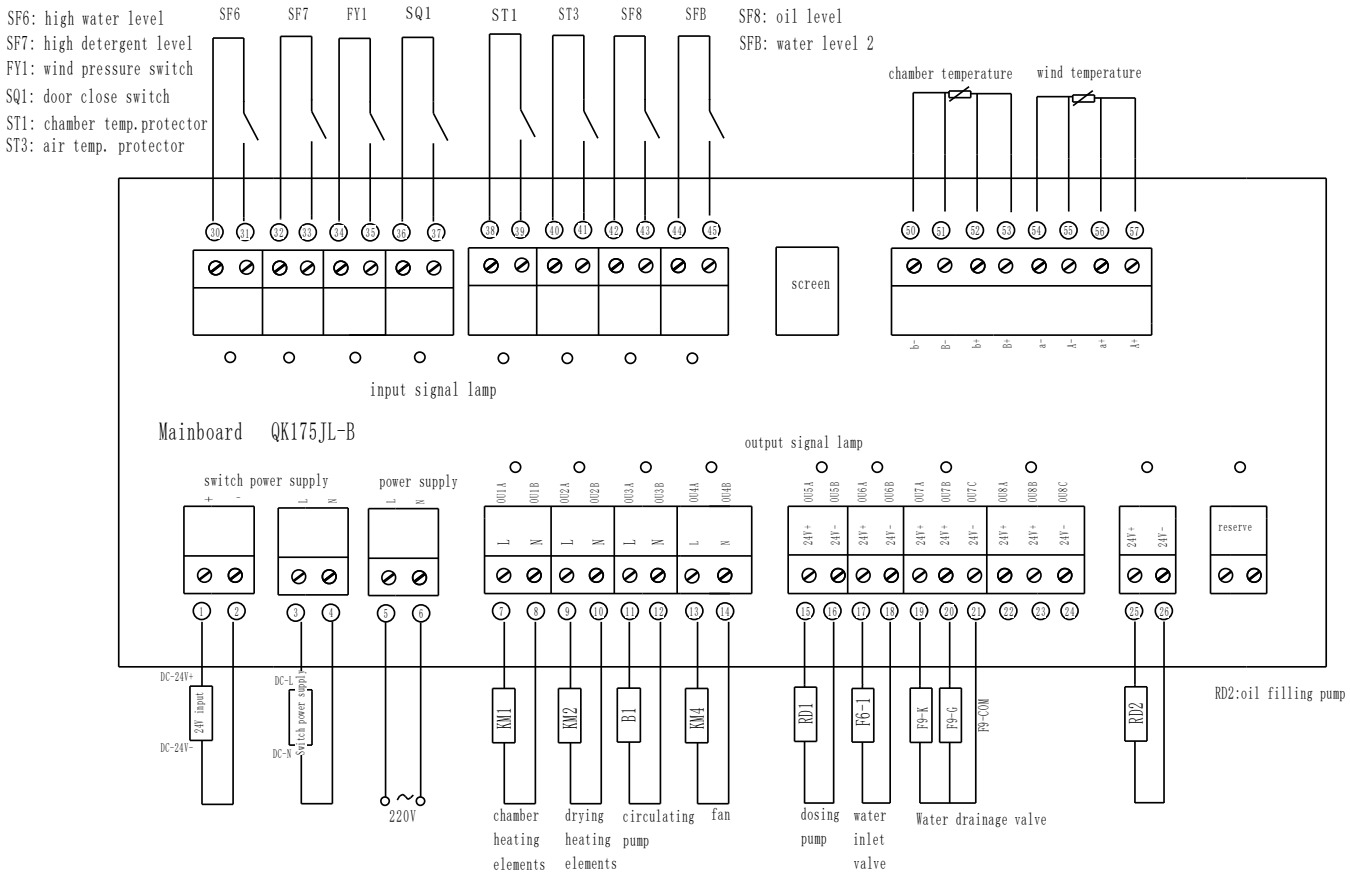
Note:

In manual operation, pay attention to the circulation pump shall not be idle (cleaning hold no water.

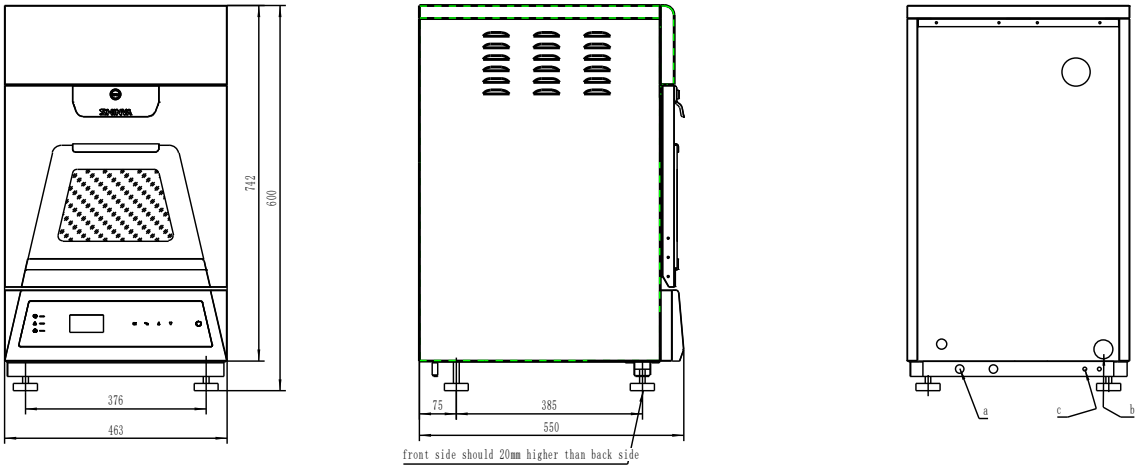
In the process of the program is run prohibited to open the door.

11.2. Drawings

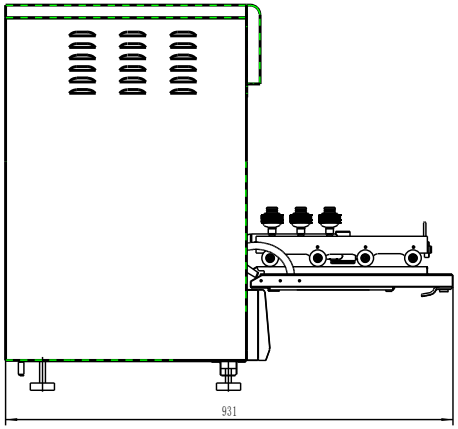
11.2.1.Electrical schematic diagram



11.2.2.installation diagram



Door opened position



Remark	a	b	c
Name	water inlet port	water drainage port	dosing port
Nozzle type	external thread	external diameter	external diameter
Nozzle size	1/2"	φ25	φ9

Technical requirements

1. The drain pipe diameter 20 mm, connect to drainage area attention!Discharge water temperature in the range of 0-93 ℃.
2. The water inlet joint is 1/2 "thread, can pick up to 20 mm silicone tube diameter, or pick up 1/2" then other materials to the joint of the pipe (pipe Canon of not less than 14 mm).
3. The water requirement: pure water, pressure is 0.2 ~ 0.2 MPa, traffic: 10 l/min, pure water ,conductivity 15 mu S/cm or less (25 ℃).
4. The power supply requirements: voltage 220 v / 50 hz, 3.5 KW,
5. The advantages of surface must level off, the place after cleaning machine without shaking, can carry under 50 kg.
6. Handling requirements:
6.1. Artificial goes on, when transporting equipment, must be two people operation is completed.
6.2. You can use the forklift.
7. The equipment weight: 50 kg/short weight.
8. The picture size for: mm

11.2.3.The equipment is mainly parameter table

Size	Chamber size/volume	Boundary dimension		Dental washing rack size	Instrument washing rack size	Net weight	Run weight
	（L×W×H）mm	（L×W×H）mm		（L×W×H）mm	（L×W×H）mm	（Kg）	（Kg）
	373*375*368	463x550x600（without dry） 463x550x750（with dry）		363*363*214	332*127*44	42	55
Performance	Disinfection temperature	Washing temperature	Disinfect time	Cycle time	Dry temperature	Dry time	Washing time
	93℃	45℃	150s	55min（with dry） 38min（no dry）	110℃	15min	300s
Program NO	0、1	0、1	0、1	0、1	0、1	0、1	0、1
MAX loading	Dental washing rack	Bottle	Conical flask	Humidification bottle	Instrument washing rack		Volume（L）
	18 handpiece	10	10	10	4 DIN basket		50
Supply	Pure water/softer water			Power			
	Pressurempa	Flow		Voltage/frequency	Power		
	0.2～0.5	25Kg/min		220/50HZ	3.5kva		
Consume	Water(standard cycle)	Clean fluid(standard cycle)		Oil（if have，standard cycle）	Electricity	寿命	
	14 L	3500ml×concentration ratio		3500ml×concentration ratio	1.2 KW/cycle	10000 times	